

## SYDNEY RICHMOND, Australia

WMO: 947530

Lat: 33.6003S

Lon: 150.7761E

Elev: 20

StdP: 101.08

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 0.1        | 1.4 | -2.9                          | 3.0 | 13.7 | -1.6 | 3.3 | 11.6 | 10.6                  | 17.1 | 9.4 | 17.2 | 0.3                   | 90   | 0.405 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 12.0                   | 36.9            | 21.1 | 34.2 | 20.9 | 32.0 | 20.4 | 23.8                | 30.8 | 23.0 | 29.3 | 22.3 | 28.1 | 5.1                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.1                            | 16.8 | 25.6 | 21.2 | 15.9 | 24.8 | 20.8 | 15.5 | 24.3 | 71.6          | 31.0 | 68.3 | 29.2 | 65.5 | 27.7 | 31.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 9.3               | 8.1  | 7.0 | -2.7                       | 41.7 | 1.1                | 2.3 | -3.5                                               | 43.3 | -4.2       | 44.7 | -4.8       | 46.0 | -5.6       | 47.7 |
|                   |      |     | -2.8                       | 26.1 | 1.1                | 1.6 | -3.7                                               | 27.3 | -4.3       | 28.2 | -5.0       | 29.0 | -5.8       | 30.2 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 17.5  | 24.1  | 23.3  | 21.4  | 17.9  | 14.0  | 11.6  | 10.7  | 11.9  | 15.2  | 18.0  | 20.4  | 22.5 | (6)  |
|      |                                                                   | DBStd       | 5.39  | 3.26  | 2.91  | 2.53  | 2.50  | 2.41  | 2.38  | 2.30  | 2.43  | 2.81  | 3.20  | 3.34  | 3.14 |      |
|      |                                                                   | HDD10.0     | 38    | 0     | 0     | 0     | 0     | 1     | 11    | 18    | 8     | 0     | 0     | 0     | 0    |      |
|      |                                                                   | HDD18.3     | 982   | 1     | 1     | 4     | 36    | 136   | 203   | 236   | 199   | 102   | 46    | 16    | 3    |      |
|      |                                                                   | CDD10.0     | 2793  | 436   | 372   | 354   | 238   | 126   | 58    | 40    | 67    | 156   | 248   | 312   | 387  |      |
|      |                                                                   | CDD18.3     | 695   | 178   | 140   | 99    | 24    | 1     | 0     | 0     | 1     | 7     | 36    | 77    | 132  |      |
|      |                                                                   | CDH23.3     | 6017  | 1536  | 1054  | 642   | 203   | 18    | 1     | 3     | 14    | 176   | 479   | 749   | 1142 |      |
| (14) | Wind                                                              | WSAvg       | 2.6   | 3.0   | 2.7   | 2.5   | 2.0   | 1.9   | 2.0   | 2.2   | 2.6   | 2.9   | 2.9   | 3.0   | 3.0  | (14) |
|      |                                                                   | PrecAvg     | 842   | 100   | 101   | 91    | 73    | 56    | 60    | 35    | 52    | 38    | 60    | 86    | 76   |      |
|      |                                                                   | PrecMax     | 1466  | 323   | 422   | 230   | 318   | 221   | 267   | 130   | 346   | 167   | 167   | 396   | 230  |      |
|      |                                                                   | PrecMin     | 397   | 7     | 7     | 4     | 3     | 4     | 2     | 0     | 0     | 3     | 0     | 10    | 2    |      |
|      |                                                                   | PrecStd     | 237   | 76    | 93    | 61    | 78    | 52    | 61    | 36    | 76    | 33    | 47    | 74    | 63   |      |
|      |                                                                   | DB          | 41.1  | 38.8  | 34.5  | 31.1  | 25.3  | 21.6  | 23.1  | 25.1  | 31.2  | 34.8  | 37.9  | 38.6  |      |      |
|      |                                                                   | MCWB        | 22.0  | 22.4  | 21.3  | 18.5  | 15.9  | 14.5  | 14.3  | 13.5  | 16.5  | 18.0  | 19.5  | 20.9  |      |      |
| (20) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | DB          | 37.0  | 34.8  | 31.1  | 27.7  | 23.1  | 19.5  | 20.0  | 22.8  | 28.0  | 31.4  | 34.0  | 35.0  |      | (20) |
|      |                                                                   | MCWB        | 22.3  | 22.4  | 21.1  | 18.3  | 15.4  | 13.3  | 12.1  | 13.2  | 15.3  | 17.6  | 19.7  | 20.8  |      |      |
|      |                                                                   | DB          | 33.8  | 31.5  | 28.9  | 25.5  | 21.4  | 18.1  | 18.1  | 20.7  | 25.1  | 28.5  | 30.1  | 32.0  |      |      |
|      |                                                                   | MCWB        | 21.8  | 21.9  | 20.6  | 17.8  | 14.8  | 12.7  | 11.7  | 12.5  | 14.8  | 17.0  | 18.8  | 20.1  |      |      |
|      |                                                                   | DB          | 30.8  | 29.0  | 26.9  | 23.8  | 20.0  | 17.0  | 16.9  | 18.9  | 22.6  | 25.8  | 27.3  | 29.2  |      |      |
|      |                                                                   | MCWB        | 21.4  | 21.1  | 20.0  | 17.2  | 14.2  | 12.2  | 11.0  | 11.6  | 14.0  | 16.4  | 18.4  | 19.6  |      |      |
|      |                                                                   | WB          | 25.3  | 25.3  | 23.8  | 21.1  | 19.0  | 16.2  | 16.0  | 18.2  | 20.6  | 22.9  | 23.7  |       |      |      |
| (28) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | MCDB        | 34.1  | 33.7  | 28.7  | 25.8  | 20.9  | 19.1  | 19.4  | 21.7  | 26.1  | 27.5  | 30.3  | 31.5  |      | (28) |
|      |                                                                   | WB          | 24.0  | 23.8  | 22.5  | 19.7  | 17.2  | 15.1  | 14.0  | 14.6  | 17.0  | 19.4  | 21.4  | 22.5  |      |      |
|      |                                                                   | MCDB        | 31.5  | 30.1  | 27.4  | 24.1  | 20.3  | 17.2  | 17.1  | 19.6  | 23.6  | 26.1  | 28.7  | 30.2  |      |      |
|      |                                                                   | WB          | 23.0  | 22.9  | 21.7  | 19.0  | 16.1  | 14.2  | 12.8  | 13.5  | 16.0  | 18.4  | 20.4  | 21.6  |      |      |
|      |                                                                   | MCDB        | 29.9  | 28.0  | 26.2  | 22.9  | 19.6  | 16.4  | 16.1  | 18.2  | 21.9  | 24.9  | 27.1  | 28.4  |      |      |
|      |                                                                   | WB          | 22.2  | 22.1  | 20.9  | 18.2  | 15.2  | 13.4  | 11.9  | 12.5  | 15.1  | 17.5  | 19.5  | 20.8  |      |      |
|      |                                                                   | MCDB        | 28.3  | 27.0  | 24.9  | 22.1  | 18.8  | 15.7  | 15.4  | 17.2  | 21.0  | 23.4  | 25.3  | 26.5  |      |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 12.0  | 10.8  | 10.6  | 11.7  | 12.9  | 11.9  | 13.3  | 14.4  | 14.2  | 13.5  | 12.2  | 12.1  |      | (35) |
|      |                                                                   | 5% DB       | 18.2  | 15.9  | 14.4  | 14.8  | 14.9  | 13.4  | 15.6  | 17.5  | 19.2  | 19.3  | 18.4  | 17.9  |      |      |
|      |                                                                   | MCWBR       | 5.7   | 5.3   | 5.6   | 6.7   | 8.0   | 8.1   | 9.0   | 9.2   | 8.5   | 7.6   | 6.6   | 5.9   |      |      |
|      |                                                                   | 5% WB       | 14.6  | 12.7  | 11.8  | 12.1  | 11.9  | 10.2  | 12.0  | 14.4  | 15.5  | 14.8  | 14.6  | 14.0  |      |      |
|      |                                                                   | MCWBR       | 5.2   | 5.0   | 5.2   | 5.9   | 7.0   | 6.8   | 7.6   | 8.2   | 7.5   | 6.7   | 6.0   | 5.5   |      |      |
|      |                                                                   | taub        | 0.397 | 0.393 | 0.363 | 0.333 | 0.316 | 0.307 | 0.298 | 0.305 | 0.337 | 0.361 | 0.389 | 0.395 |      |      |
|      |                                                                   | taud        | 2.430 | 2.464 | 2.529 | 2.585 | 2.580 | 2.607 | 2.614 | 2.585 | 2.492 | 2.453 | 2.417 | 2.416 |      |      |
| (42) | Clear-Sky Solar Irradiance                                        | Ebn at Noon | 941   | 924   | 917   | 892   | 856   | 840   | 870   | 911   | 928   | 944   | 943   | 947   |      | (42) |
|      |                                                                   | Edh at Noon | 122   | 114   | 100   | 84    | 75    | 69    | 71    | 82    | 102   | 114   | 123   | 125   |      |      |
|      |                                                                   | RadAvg      | 6.39  | 5.47  | 4.68  | 3.82  | 3.06  | 2.42  | 2.84  | 3.74  | 4.95  | 5.69  | 6.00  | 6.58  |      |      |
| (44) | All-Sky Solar Radiation                                           | RadStd      | 0.50  | 0.52  | 0.43  | 0.33  | 0.22  | 0.21  | 0.21  | 0.23  | 0.29  | 0.57  | 0.57  | 0.64  |      | (44) |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only            |  | N/A   | N/A     | N/A    | N/A     | +0.63 | N/A   | N/A         | N/A     | +206    | +155    |
| Regional (37 neighbors) |  | +0.44 | N/A     | N/A    | +0.77   | N/A   | N/A   | -2          | -86     | +185    | +99     |

Nomenclature: See separate page